

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**First Semester of B. Pharm Examination****University Theory Examination January 2018****MA131.1 Basic Mathematics**

Date: 01/01/2018, Monday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION -I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION-I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION-I is 30 minutes. Answers to SECTION-I must be written during the first 30 minutes of the examination.
3. Answers to SECTION-II and SECTION-III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.
6. Use of scientific non-programable calculator is allowed.

SECTION I**Q - 1** Choose the correct answer for the following questions each .**[20]**

1. If E and F are two sets then which of the following set relation is not true?
 (A) $E \cup F = \{x|x \in E \text{ or } x \in F\}$ (C) $E \cap F' = \{x|x \in E, x \notin F\}$
 (B) $E \cap F = \{x|x \in E, x \in F\}$ (D) $E \cap F' = \{x|x \notin E, x \in F\}$
2. Consider the sets $U = \{1, 2, \dots, 9\}$, $E = \{1, 2, 3, 4\}$, $F = \{2, 4, 6, 8\}$ and $G = \{3, 4, 5, 6\}$. Then $(E \cup F)' =$ _____
 (A) $\{3, 4\}$ (B) $\{2, 8\}$ (C) $\{5, 7, 9\}$ (D) ϕ
3. Suppose $n(U) = 110$, $n(E) = 30$, $n(F) = 35$ and $n(E \cap F) = 20$ then $n(E \cap F') =$ _____
 (A) 10 (B) 80 (C) 15 (D) 90
4. Let $E = \{1, 2\}$, and $F = \{a, b, c\}$ then $n(E \times F) =$ _____
 (A) 4 (B) 9 (C) 6 (D) 5
5. Consider the relations f, g and h on the set $D = \{1, 2, 3\}$ given by $f = \{(1, 3), (2, 3), (3, 1)\}$, $g = \{(1, 2), (3, 1)\}$ and $h = \{(1, 3), (2, 1), (1, 2), (3, 1)\}$ then,
 (A) f is not function from D into D . (C) g is function from D into D .
 (B) f is function from D into D . (D) h is function from D into D .
6. Let the function $f: \mathcal{R} \rightarrow \mathcal{R}$ be defined by $f(x) = x^2 - 3x + 2$ then $f(-3) =$ _____
 (A) 2 (B) 20 (C) -9 (D) 9
7. The value of $\log_2 8 \times \log_2 16 =$ _____
 (A) 12 (B) 4 (C) 128 (D) 7

8. Let $V = \{-1, 0, 1, 2\}$. Let the function $g : V \rightarrow \mathcal{R}$ given by $g(x) = x^2 + 1$, then the range of $g =$ _____
 (A) $\{1, 2\}$ (B) $\{-1, 0, 1, 2\}$ (C) $\{1, 2, 5\}$ (D) $\{0, 1, 2\}$
9. Let $E = \{-1, 0, 1\}$ and let the functions f and g of E into E given by $f(x) = x^2, g(x) = x^3$. Then
 (A) g is onto function. (C) g is not onto function.
 (B) f is onto function. (D) f is one-one function.
10. Given that $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} g(x) = 2$, then $\lim_{x \rightarrow a} \frac{2f(x) + 1}{1 - f(x)g(x)} =$ _____
 (A) -3 (B) 3 (C) 1 (D) 2
11. $\lim_{x \rightarrow 2} x(x-1)(x+1) =$ _____
 (A) 0 (B) 2 (C) 3 (D) 6
12. The slope of the curve $y = 1 - x^2$ at point $x = 2$ is _____
 (A) 2 (B) -4 (C) -3 (D) 4
13. If $y = x^4 + 12x$ then derivative $\frac{d}{dx}(x^4 + 12x) =$ _____
 (A) $4x^3 + 12$ (B) $x^4 + 12$ (C) $4x^3 + 12x$ (D) $4x^3$
14. If $y = \frac{1}{x}$ then derivative $\frac{d}{dx}\left(\frac{1}{x}\right) =$ _____
 (A) $\frac{1}{x^2}$ (B) $\log x$ (C) $-\frac{1}{x^2}$ (D) $-\frac{1}{x}$
15. The integration of the function $\int 2x dx =$ _____
 (A) x^2 (B) 2 (C) $\frac{x^2}{2} + C$ (D) $x^2 + C$
16. If $P = \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}$, then $3P =$ _____
 (A) $\begin{bmatrix} 3 & -6 \\ 6 & 9 \end{bmatrix}$ (B) $\begin{bmatrix} 3 & -2 \\ 2 & 9 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & -6 \\ 6 & 3 \end{bmatrix}$ (D) $\begin{bmatrix} 3 & -6 \\ 2 & 3 \end{bmatrix}$
17. If $U = [u_{ij}] = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$ then $u_{23} =$ _____
 (A) -1 (B) 2 (C) 0 (D) 1
18. If $P = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ and $Q = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$ then $PQ =$ _____
 (A) $\begin{bmatrix} 4 & 3 \\ 6 & 6 \end{bmatrix}$ (B) $\begin{bmatrix} 4 & 2 \\ 6 & 4 \end{bmatrix}$ (C) $\begin{bmatrix} 4 & 2 \\ 9 & 6 \end{bmatrix}$ (D) $\begin{bmatrix} 6 & 9 \\ 3 & 6 \end{bmatrix}$

19. The degree of polynomial $(x^2 - x + 1)(x + 1)$ is _____

- (A) 2 (B) 3 (C) 1 (D) 0

20. The system of equations $x_1 + x_2 = 1, x_1 + 2x_2 = 2$ has unique solution using Cramer's rule, if

- (A) $\begin{vmatrix} 1 & -1 \\ 1 & -2 \end{vmatrix} \neq 0$ (B) $\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} \neq 0$ (C) $\begin{vmatrix} -1 & 1 \\ -2 & 1 \end{vmatrix} \neq 0$ (D) $\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} \neq 0$

SECTION II

Q - 2 Attempt any Four of the following.

[20]

A. Consider the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and the sets $E = \{1, 2, 3, 4, 5\}$, $F = \{4, 5, 6, 7\}$, $G = \{5, 6, 7, 8, 9\}$. Find the set relations and construct venn-diagrams of each.

[05]

(a) $E \cup F$ and $E \cap F$

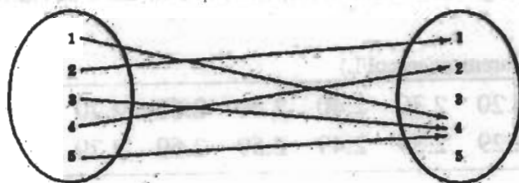
(c) $E \cap F'$

(b) $E \cup G$ and $E \cap G$

(d) $E' \cap F$

B. Consider the function $f : A \rightarrow B$ defined by the diagram. Find

[05]



- (a) $f^{-1}(2)$ (b) $f^{-1}(3)$ (c) $f^{-1}(4)$
(d) $f^{-1}\{1, 2\}$ (e) $f^{-1}\{2, 3, 4\}$

C. Let $E = \{1, 2\}$, $F = \{a, b, c\}$ and $G = \{c, d\}$ then Find

[05]

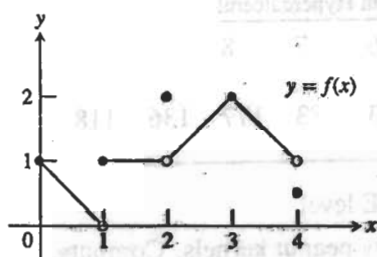
(a) $(E \times F) \cap (E \times G)$ (b) $E \times (F \cap G)$

D. Evaluate the limits (a) $\lim_{x \rightarrow 5} \frac{x-5}{x^2-25}$ (b) $\lim_{x \rightarrow -1} \frac{x^3+4x^2-3}{x^2+5}$

[05]

E. Consider the function $y = f(x)$ in figure whose domain is the closed interval $[0, 4]$

[05]



Discuss the continuity of f at $x = 0, 1, 2, 3$, and 4 .

F. Evaluate the limit $\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1}$

[05]

SECTION III

Q - 3 Attempt any Four of the following.

[20]

A. Evaluate the integral $\int \frac{4+\sqrt{t}}{t^3} dt$

[05]

B. Evaluate the integral $\int_0^3 (3x^2 + x - 5) dx$

[05]

- C. If $M = \begin{bmatrix} 3 & 6 \\ 4 & 7 \end{bmatrix}$ and $N = \begin{bmatrix} 8 & -3 \\ 5 & -6 \end{bmatrix}$
Find (a) $|M|$ (b) $|N|$ (c) show that $|MN| = |M||N|$ [05]

- D. Find the derivative of $y = \frac{(x-1)(x^2-2x)}{x^4}$ [05]

- E. If $P = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ and $Q = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ then compute
(a) $2P + 3Q$ (b) $3P - 4Q$ [05]

- F. Find the first and second derivative of $y = \frac{x^3}{3} + \frac{x^2}{2} + \frac{x}{4}$ [05]

Q - 4 Attempt any Four of the following. [20]

- A. Solve the following system of equations
 $x + y + z = 7, x + 2y + 3z = 16, x + 3y + 4z = 22$ using Cramer's rule. [05]

- B. The following table (Table 1) represents frequency distribution of measurement of calcium (mmol/L) of subjects 65 years of age and older. Compute the standard deviation of calcium measurement. [05]

Table 1: Calcium Measurement (mmol/L)

Lower Limit	1.90	2.00	2.10	2.20	2.30	2.40	2.50	2.60	2.70
Upper Limit	1.99	2.09	2.19	2.29	2.39	2.49	2.59	2.69	2.79
Number of Subjects	1	2	16	36	50	43	21	6	2

- C. A paper by Robertson et al. [1976] discusses the level of plasma prostaglandin E (iPGE) in patients with cancer with and without hypercalcemia. (Table 2)

Table 2: Plasma Prostaglandin E Levels: Patients with Hypercalcemia

Patient Number	1	2	3	4	5	6	7	8	9	10
Plasma iPGE (pg/ml)	500	500	301	272	226	183	183	177	136	118

Compute mean and standard deviation of plasma iPGE level. [05]

- D. The following data deal with the aflatoxin levels of raw peanut kernels. Compute,
(a) All quartiles (b) Quartile deviation [05]

Table 3: aflatoxin levels

30	26	26	36	48	50	16	31	22	27	23	35	52	28	37
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- E. (a) Solve the system of equations $x_1 + x_2 = 1, x_1 + 2x_2 = 2$ using Cramer's rule. [03]

- (b) Solve the quadratic equation $x^2 + 5x + 6 = 0$ [02]

- F. Consider data (Table 1) of question B. Compute the Mode of calcium measurement. [05]